Robustness of Vision Transformers Under Image Corruption

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Track B - LVM Applications (Vision)

1. Introduction & Motivation

Deep learning models for image classification have achieved amazing accuracy on standard benchmarks, but high benchmark performance does not necessarily translate to reliability in real-world deployments. In practice, images captured by cameras are likely to face a wide range of natural corruption. Things that could range from noise from low-light sensors, blur from motion or defocus; brightness shifts from lighting changes, and geometric distortions from camera angles. A model that performs well on clean test images but degrades severely under these conditions poses serious risks in safety-critical applications such as autonomous driving, medical imaging, and surveillance systems.

Vision Transformers (ViT) have emerged as a powerful alternative and competitor to Convolutional Neural Networks (CNNs) for image classification, achieving state-of-the-art results on many levels. Unlike CNNs, which rely on local convolutional filters with strong spatial inductive biases, ViTs process images as sequences of patches using global self-attention mechanisms. This architectural difference raises an important and underexplored question: does ViT's global attention make it more or less robust to image corruptions compared to CNN's local feature extraction?

This project compares how well a Vision Transformer (ViT-Base) and a CNN (ResNet-18) hold up when test images are corrupted, using four corruption types at three intensity levels on the CIFAR-10 dataset.

Success Criterion: Successfully train both models to strong clean-image baselines, apply controlled corruptions to the test set only, quantify per-corruption accuracy degradation, and produce a clear comparative robustness analysis without requiring either model to improve, only to measure and analyze.

2. Related Work

The robustness of deep learning models to image corruption has been studied a lot in the context of CNNs. Hendrycks and Dietterich (2019) introduced ImageNet-C, a standardized benchmark of 75 corruption types, showing that standard CNN models degrade significantly under distribution shifts. Their work established accuracy drops and mean corruption errors as standard robustness metrics. Since Vision Transformers were introduced by Dosovitskiy et al. (2020), researchers have been studying how robust they are compared to CNNs. Zhou et al. (2022) found that ViTs tend to focus on object shape while CNNs focus more on texture. Using attention map visualizations, they showed that the two architectures pick up on different visual features, which means they also fail in different ways when images are corrupted.

Karkala et al. (2025) specifically measured ViT robustness under realistic corruptions, finding that ViTs generally outperform CNNs on geometric and photometric corruptions but show comparable or worse performance under severe pixel-level noise. Our work builds directly on this finding by testing it on the CIFAR-10 benchmark with controlled severity levels, providing fine-grained analysis of which corruption types and severity levels drive the performance gap.

3. Data Statement

3.1 Dataset

We use CIFAR-10 (Krizhevsky, 2009), a standard image classification benchmark consisting of 60,000 color images of size 32x32 pixels across 10 classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, and truck. The dataset is balanced with exactly 1,000 images per class in the test set and 5,000 per class in the training set.

- Source: University of Toronto, Alex Krizhevsky (2009)
- License: Publicly available for academic research use with no commercial restrictions
- Official download: https://www.cs.toronto.edu/~kriz/cifar.html
- Size: 60,000 images total: 50,000 train (5,000 from this was taken forvalidation) and 10,000 test

3.2 Split Plan

We follow a fixed train/validation/test split using random seed 42 for full reproducibility:

Split	Size	Purpose
Train	45,000 images	Model training with augmentation
Validation	5,000 images (held out from train)	Hyperparameter tuning, early stopping
Test (clean)	10,000 images	Final clean accuracy evaluation
Test (corrupted)	10,000 images per corruption/severity	Robustness evaluation only

3.3 Preprocessing

Training augmentations (applied to training set only): random crop with 4-pixel padding, random horizontal flip, normalization with CIFAR-10 mean (0.4914, 0.4822, 0.4465) and standard deviation (0.2023, 0.1994, 0.2010). For ViT, images are additionally resized to 224x224 to match the pretrained model's expected input size and normalized with mean/std of 0.5 for all channels. Test-time corruptions are applied after converting images back to the [0,1] range, prior to model-specific normalization. This ensures corruptions are applied in a consistent, architecture-agnostic manner.

3.4 Leakage Risks & Mitigations

- Corruptions are applied exclusively at evaluation time, the training set is always clean
- No hyperparameter tuning was performed on the corrupted test set
- Train/validation split uses a fixed seed (42) documented in all config files
- No external data or pre-corrupted datasets were used

4. Baseline

We use ResNet-18 pretrained on ImageNet as our CNN baseline. We decided that ResNet-18 is an appropriate baseline for this study for several reasons: it is a well-established, widely benchmarked architecture with strong inductive biases for image classification; its local convolutional filters represent a generic CNN approach to visual feature extraction; and it achieves competitive accuracy on CIFAR-10

with moderate compute requirements, making it a practical comparison point. The pretrained ResNet-18 was fine-tuned on CIFAR-10 by replacing the final fully connected layer with a 10-class output head. This gives the baseline the benefit of ImageNet pretraining, ensuring the comparison to ViT (which also uses ImageNet pretrained weights) is fair.

Baseline clean test accuracy: 83.84%

5. Method

5.1 Model Architecture

Our primary model is ViT-Base/16 (google/vit-base-patch16-224), a Vision Transformer pretrained on ImageNet-21k and fine-tuned on ImageNet. ViT divides each input image into fixed-size 16x16 patches, linearly embeds them, and processes them through 12 transformer encoder blocks using multi-head self-attention. This global attention mechanism contrasts directly with ResNet-18's local convolutional filters, making the comparison architecturally meaningful for studying robustness.

Property	CNN (ResNet-18)	ViT (ViT-Base/16)
Feature extraction	Local convolution filters	Global self-attention on patches
Inductive bias	Strong (translation equivariance)	Weak (learned from data)
Pretrained on	ImageNet-1k	ImageNet-21k + ImageNet-1k
Input size	32x32 (native)	224x224 (upscaled)
Parameters	~11M	~86M
Clean Test Accuracy	83.84%	97.88%

5.2 Fine-tuning Strategy

For ViT, we fine-tune only the classifier head and the last two transformer encoder blocks (layers 10 and 11), keeping earlier layers frozen. This partial fine-tuning approach was chosen because we were adapting large pretrained vision models to smaller datasets. So partial fine tuning helped prevent any overfitting to occur while allowing the model to adapt its high-level representations to CIFAR-10's class structure. Full fine-tuning of all 86M parameters on a 45k-image dataset would more likely cause it to have overfitting and instability. Though for ResNet-18, all layers have been fine-tuned from the pretrained ImageNet weights, which is appropriate given its smaller parameter count (~11M) and stronger inductive biases.

5.3 Corruption Methodology

We evaluated robustness under four corruption types at three severity levels each, applied only to the test set. These corruptions were chosen to simulate realistic real-world image degradations:

Corruption Type	Severity 1	Severity 2	Severity 3
Gaussian Noise	$\sigma = 0.25$	$\sigma = 0.50$	$\sigma = 0.75$
Gaussian Blur	$\sigma = 1.0$	$\sigma = 2.0$	$\sigma = 4.0$

Corruption Type	Severity 1	Severity 2	Severity 3
Brightness (dimming)	scale = 0.5	scale = 0.3	scale = 0.1
Rotation	15°	30°	45°

All corruptions are implemented using PyTorch and SciPy, applied to raw [0,1] pixel tensors before model-specific normalization. Each corruption level is evaluated independently on the full 10,000-image test set.

6. Training & Compute

Setting	CNN (ResNet-18)	ViT-Base/16
Optimizer	Adam	Adam
Learning rate	1e-4	2e-5
Weight decay	1e-4	1e-4
Batch size	128	64
Epochs	15	10
LR schedule	Cosine annealing	Cosine annealing
Fine-tuned layers	All layers	Classifier + last 2 blocks
Hardware	Kaggle T4 x2 GPU	Kaggle T4 x2 GPU
Approx. runtime	~30 minutes	~90 minutes
Random seed	42	42

Code development and collaboration were done through Google Colab, which provided a shared environment for the team to work together. Model training and evaluation were run on Kaggle Notebooks using a T4 x2 GPU configuration, as Kaggle offered more reliable GPU access within our free-tier compute budget. A learning rate schedule that gradually reduces the learning rate over training was used for both models to be able to convergence more stably. Early stopping based on validation accuracy was implemented to prevent overfitting.

7. Experiments & Evaluation

7.1 Primary Metrics

- Top-1 Accuracy: proportion of correctly classified test images
- Accuracy Drop (%): difference between clean and corrupted accuracy for each model
- Robustness Gap: difference in accuracy between ViT and CNN under each corruption (positive = ViT more robust)

7.2 Clean Accuracy Baseline

Before evaluating robustness, we establish strong clean-image baselines to ensure that accuracy drops under corruption are meaningful:

Model	Clean Val Accuracy	Clean Test Accuracy
CNN (ResNet-18)	82.96%	83.84%
ViT-Base/16	97.78%	97.88%

ViT achieves substantially higher clean accuracy (97.88% vs 83.84%), which is expected given its larger parameter count and stronger pretraining. This gap is important context for interpreting robustness results, ViT starts from a higher baseline, so even larger absolute drops can still leave it ahead of CNN.

7.3 Robustness Results

The following table summarizes Top-1 accuracy for both models across all corruption types and severity levels, along with the robustness gap (positive values indicate ViT is more robust):

Corruption	Severity	CNN Acc	ViT Acc	Gap (ViT - CNN)
Gaussian Noise	1 (σ=0.25)	26.7%	21.8%	-4.8%
Gaussian Noise	2 (σ=0.50)	13.2%	10.2%	-3.0%
Gaussian Noise	3 (σ=0.75)	10.9%	10.0%	-0.9%
Gaussian Blur	1 (σ=1.0)	53.9%	77.6%	+23.6%
Gaussian Blur	2 (σ=2.0)	29.9%	30.9%	+1.0%
Gaussian Blur	3 (σ=4.0)	23.5%	16.4%	-7.1%
Brightness	1 (scale=0.5)	77.0%	96.0%	+19.1%
Brightness	2 (scale=0.3)	62.0%	89.4%	+27.4%
Brightness	3 (scale=0.1)	14.2%	30.3%	+16.1%
Rotation	1 (15°)	68.2%	92.2%	+24.1%
Rotation	2 (30°)	46.1%	84.7%	+38.5%
Rotation	3 (45°)	35.8%	72.2%	+36.4%

7.4 Key Findings

Finding 1: Gaussian Noise: Models are equally vulnerable.

Both models collapse to near-random performance under heavy Gaussian noise (σ=0.75). CNN actually performs slightly better than ViT at severity levels 1 and 2, though both are well below useful accuracy. This is consistent with prior work suggesting that ViT's patch-based tokenization is particularly sensitive to pixel-level noise, since noise corrupts individual patch embeddings that the attention mechanism then propagates globally. CIFAR-10's small 32x32 image size amplifies this effect; noise at σ=0.25 already obscures a significant fraction of the signal.

Finding 2: Gaussian Blur: ViT is more robust at mild and moderate blur.

At severity 1, ViT outperforms CNN by 23.6 percentage points (77.6% vs 53.9%). However, this advantage reverses at severity 3, where CNN slightly outperforms ViT (23.5% vs 16.4%). This suggests ViT's global attention can leverage low-frequency shape information preserved in mildly blurred images, but severe blur destroys the patch-level structure ViT relies on.

Finding 3: Brightness: ViT is dramatically more robust across all severity levels.

ViT maintains 96.0%, 89.4%, and 30.3% accuracy at the three brightness severity levels, compared to CNN's 77.0%, 62.0%, and 14.2%. The robustness gap is largest at severity 2 (+27.4%). ViT's global attention appears better suited to extracting shape and structural features that remain informative even under significant brightness reduction, while CNN's local filters may rely more heavily on absolute pixel intensity values.

Finding 4: Rotation: ViT's strongest advantage, especially at high severity.

This is the most surprising finding. At 45° rotation, ViT achieves 72.2% accuracy while CNN drops to 35.8%, a gap of 36.4 percentage points. ViT maintains over 70% accuracy even at the most severe rotation, while CNN degrades to below chance level for some classes. CNN's translation equivariance does not extend to rotations, and its local filters are sensitive to orientation. ViT's global self-attention, having no built-in geometric inductive bias, appears to learn more rotation-invariant representations from the pretrained ImageNet data.

7.5 Error Analysis

Confusion matrix analysis under Gaussian noise (severity 3) reveals that both models show significant class confusion, though with different patterns. CNN disproportionately predicts 'frog' across many true classes: frog images in CIFAR-10 tend to have distinctive green color distributions that may provide a robust signal even under heavy noise. ViT's errors are more spread across semantically similar classes (cat/dog, deer/horse), suggesting it relies more on shape and structural features that become ambiguous under noise. Per-class analysis shows that classes with distinctive shape profiles (frog, truck) are more robust under CNN, while classes requiring global context (cat, deer) are more robust under ViT. This is consistent with the architectural difference between local filter-based and global attention-based feature extraction.

7.6 Ablation

We treat the three severity levels as an ablation study of corruption intensity. For rotation, ViT's accuracy degrades slowly (92.2% --> 84.7% --> 72.2%) while CNN degrades rapidly (68.2% --> 46.1% --> 35.8%), demonstrating that ViT's advantage increases with corruption severity. For Gaussian noise, both models degrade rapidly and converge near random performance at severity 3, confirming noise as a universal failure mode for both architectures at the 32x32 image scale.

8. Responsible AI & Limitations

8.1 Dataset Bias

CIFAR-10 is a simplified, balanced benchmark dataset that does not represent the full diversity of real-world images. All images are 32x32 pixels, which is significantly smaller than images in real-world deployment scenarios. Robustness conclusions drawn from CIFAR-10 may not directly transfer to larger-scale or more diverse datasets such as ImageNet or domain-specific medical/satellite imagery. The dataset contains no demographic or personally identifiable information.

8.2 No Personal Data

This project uses exclusively the publicly available CIFAR-10 dataset. No personal, private, or sensitive data was collected, used, or stored at any point. The dataset contains images of objects and animals only.

8.3 Robustness Failure Risks

Our results demonstrate that both models can fail severely under realistic corruptions. CNN accuracy drops to 10.9% under heavy Gaussian noise, and ViT drops to 10.0%, both near random performance for a 10-class problem. Deploying either model in safety-critical systems (autonomous vehicles, medical diagnostics) without corruption-robust training or input preprocessing would be inadvisable. The rotation robustness gap is particularly important for applications where camera orientation cannot be controlled.

8.4 Transparency & Mitigation

- All accuracy degradation results are reported transparently, including cases where ViT underperforms CNN
- This report includes all 12 conditions without any kind of filtering.
- AI tools (Claude, Anthropic) were used to assist with code structure and debugging. All notebook building, experimental results, analysis, and writing represent the team's own work
- To mitigate these errors in any future work, we will include more advanced training (e.g., AugMix, DeepAugment) and allow it to adjust itself while it's being used.

9. Reproducibility

Important Note: GitHub was not used often since Google Colab allowed us to collaborate on the same notebook by sharing it with one another and editing the code on the notebook itself and meeting in person and working on 1 device together. Though a Github was created to display and organize everything needed to reproduce what we worked on.

The Google drive was used to store all the notebooks, checkpoints, and results.

GitHub Repository**:** https://github.com/saravanan02016821/ViT-Robustness.git

Google Drive: https://drive.google.com/drive/folders/1bpb-aRyqiTvootc9Z3-V56J81wOeRRVF?usp=sharing

9.1 Environment

- Python 3.12, PyTorch 2.x, torchvision, transformers 4.35+, scipy, scikit-learn, matplotlib, seaborn
- Hardware: Kaggle Notebooks, NVIDIA T4 x2 GPU
- Full dependency list in requirements.txt in the repository

9.2 Reproducing Results

To reproduce all results from scratch:

- Clone the repository and install dependencies: pip install -r requirements.txt
- Run training notebook (02_Train_Models.ipynb) on Kaggle with T4 GPU, CNN ~30 min, ViT ~2.5-4.5 hours

- Run evaluation notebook (03_Evaluation.ipynb) with saved checkpoints: ~40 min on GPU
- Run plotting notebook (04_Plots_and_Analysis.ipynb) in Colab or locally: ~2 min
- All experiments use random seed 42 set via torch.manual_seed(42) and numpy.random.seed(42)
- Saved model checkpoints available at: https://drive.google.com/drive/folders/1bpb-aRyqiTvootc9Z3-V56J81wOeRRVF?usp=sharing
- CNN training: 15 epochs, batch size 128, learning rate 0.0001, learning rate gradually reduced over training
- ViT training: 10 epochs, batch size 64, learning rate 0.00002, only the last 2 layers and the final classification layer were updated during training

10. Conclusion & Future Work

10.1 Key Takeaways

This study provides a systematic comparison of ViT and CNN robustness under image corruption on CIFAR-10. Our main findings are:

- ViT is substantially more robust than CNN on 3 of 4 corruption types: Gaussian blur (mild/moderate), brightness reduction, and rotation, with the largest advantage on rotation (up to +38.5 percentage points at 45°)
- Both models are equally vulnerable to Gaussian noise, with CNN slightly outperforming ViT at low/medium noise severity, likely due to CIFAR-10's small image size amplifying noise effects on patch-based tokenization
- ViT's advantage increases with corruption severity for rotation and brightness, suggesting its global attention mechanism learns more invariant representations for geometric and photometric transformations
- Starting from a much higher clean accuracy (97.88% vs 83.84%), ViT maintains useful accuracy levels across more corruption conditions than CNN

10.2 Limitations

- CIFAR-10's 32x32 image size is atypically small, likely amplifying noise sensitivity and limiting the generalizability of conclusions to real-world image scales
- ViT was partially fine-tuned (last 2 blocks only) while CNN was fully fine-tuned, a fully fine-tuned ViT might show different robustness characteristics
- We evaluate only 4 corruption types; a more comprehensive study would include weather corruptions, compression artifacts, and adversarial perturbations
- No corruption-aware training was used, both models were trained only on clean data

10.3 Future Work

- Increase the number of corruptions each model is tested on
- Evaluate on larger-scale datasets (ImageNet-C) to confirm findings at realistic image resolutions
- Apply augmentation-based robustness training (AugMix, DeepAugment) and measure improvement
- Extend comparison to include DeiT, Swin Transformer, and EfficientNet for a broader architectural study

- Investigate test-time adaptation techniques for improving robustness without retraining

11. Contribution Log

Sharvani Bhavanam: Data & Setup

I handled downloading and setting up the CIFAR-10 dataset, wrote the code for all four corruption functions (Gaussian noise, blur, brightness, and rotation), and made sure the train/val/test split stayed fixed using seed 42 across all notebooks. I also checked and made sure that corrupted images were never mixed into the training data. I also worked on the data statement, methods and architecture, and ensuring the reproducibility on the report.

Shruthi Saravanan: Model & Training

set up and trained both models, the ResNet-18 CNN baseline and the ViT-Base/16. I configured the training settings for both, tracked performance during training, and saved the best checkpoints. Google's T4 GPU went very slowly when I tried to complete the training which led us to run out of our GPU limit so we moved our work from Google Colab to Kaggle, which is where we did all of our final training. I documented all of the data gathered during training and computing, introduction, related works that we found, on the final report as well as the limitation, future work and the AI responsibility measures taken by the group during the entire project.

Charan Sai Kamsani: Evaluation & Reproducibility

I built the evaluation pipeline that tested both models under all 12 corruption conditions and generated all the plots used in the report and presentation, including the degradation curves, confusion matrices, and per-class accuracy charts on the google colab Notebook. I also wrote what the experiments and evaluation were on the final report.

References

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